Exp: 3a UNDERSTANDING THE CISCO PACKET
23/7/26 TRACER ENVIRONMENT

Aim:

To understand the cisco Packet tracer environment and design a simple computer network using its interface and networking components

Steps:

Open the cisco Packet tracer application

Observe the user interface, including the menu bar, Main tool bar, workspace, the menu bar, Main toolbar, workspace, common tool bar, network component box, and realtime / Sim / Simulation tabs

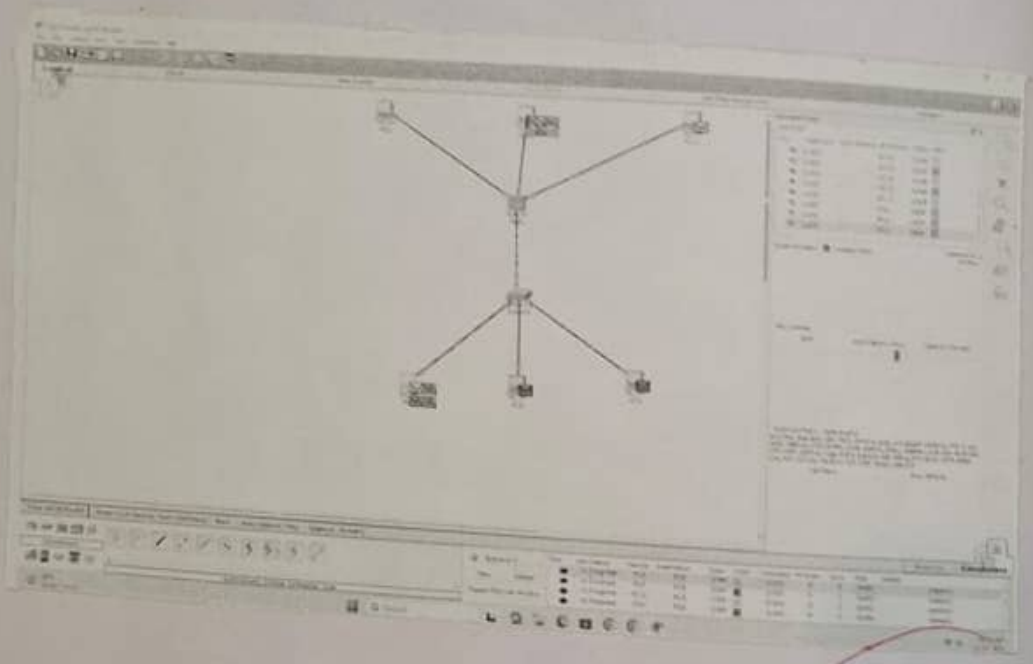
From the network component box, Select the required network devices such as PCs, hubs and switch

Drag and drop the devices using copper straight - through cable

Configure IP address for the PCs using the desktop → IP configuration option

verify that all links are active (green led)

Use the add simple PDU tool to send packet between two PCs.



Observe packet transmission in real time mode and simulation mode

Save the network topology

RESULT:

The Cisco Packet Tracer environment was successfully explored, and a simple network was designed. The internet connection, IP configuration and Packet transmission were successfully understood using the real time simulation mode

Exp: 3b1
26/7/26

HUB

Aim:-

To study the behaviour of a hub by observing how it forward data packet using cisco Packet tracer

STEPS:

(i) Open cisco Packet tracer

(ii) Drag and Place 1 Hub and 4 generic PCs into the workspace

(iii) Connecting all PCs to the Hub using copper straight-through cable

(iv) Configure IP address for all PCs in same network

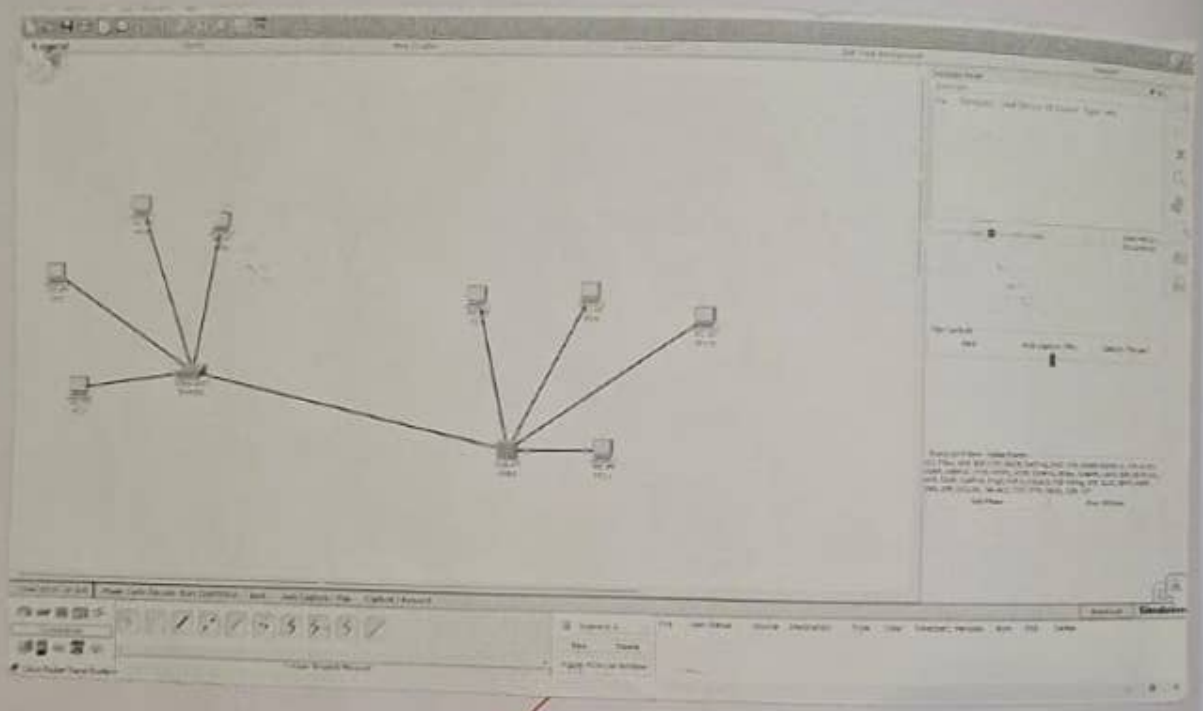
(v) Select add simple PDU from the common tools bar

(vi) Send a Packet from one PC(source) to another PC (destination)

(v) Switch to simulation mode and observe how the hub forward the Packet

Result:

The hub broadcast the received Packet to all connected device. Every PC receive the Packet, but only the destination PC accepts it while the other discard it.



Exp: 3b2
26/7/24

SWITCH

Aim:

To study the behaviour of a switch by observing how it forwards data packet using Cisco Packet Tracer

STEPS:

(i) Open Cisco Packet Tracer

(ii) Drag and place 1 switch and 4 generic PCs into the workspace

(iii) Connect all PCs to the switch using copper straight - through cables

(iv) Configure IP address for all PCs in the same network

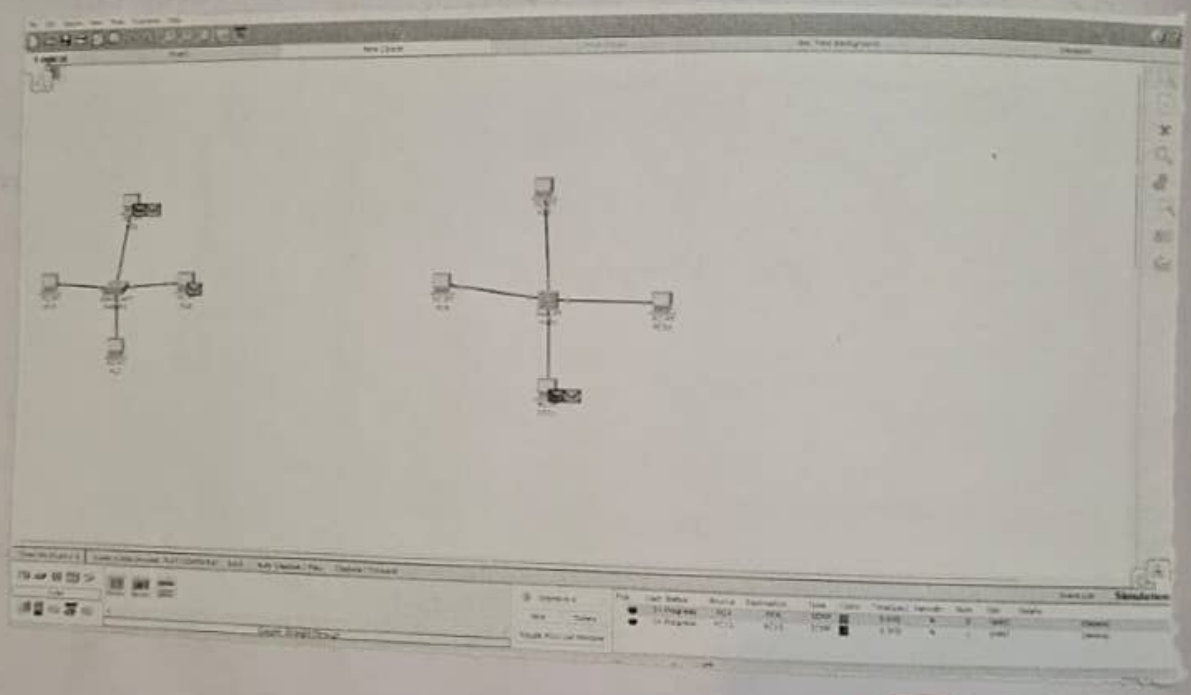
(v) Select add simple PDU from the common tool bar

(vi) Send a Packet from one PC to another PC

(vii) Switch to simulation mode and observe how the switch forward the Packet.

Result:

The switch forward the Packet only to the destination PC by using it MAC address table. This makes unnecessary network traffic and improves communication



Exp : 3b3
26/2/26

HUB AND SWITCH

Aim

To compare the packet forward behaviour of hub and a switch using cisco packet tracer

STEPS

① Create one network using a hub and another using a switch, each connected to 4 PCs

② Configure IP address for all PC in both networks

③ Send a simple PDU between two PCs in each networks

④ Observe packet transmission in simulation mode

⑤ Compare how the hub and switch forward packets

Result:

The hub broadcasting packet to all connected devices, whereas the switch forwards packet only to the intended destination Switch provides better performance and efficient handling of network